

Exponential Shot-Noise Ornstein–Uhlenbeck Dynamics: Exact Semigroup, Gamma Equilibrium, and Polynomial Filtrations

Route-A source-local certificate HCS-C335

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Abstract

We close the positive Ornstein–Uhlenbeck process driven by Poisson arrivals with exponential marks. Its all-time transition Laplace transform follows directly from the marked-Poisson functional. The same formula yields the unique Gamma equilibrium, sharp Wasserstein contraction, all stationary moments and cumulants, and the exact covariance. We further determine the generator spectrum on every finite polynomial filtration, while explicitly making no claim about the full L^2 spectrum, and close every zero-parameter boundary.

1 Frozen process and semigroup

Fix $\gamma, \kappa, \beta > 0$. Let N_t be Poisson of rate κ , let T_j be its jump times, and let the independent marks Y_j have exponential rate β . On $[0, \infty)$ consider

$$dX_t = -\gamma X_t dt + dJ_t, \quad J_t = \sum_{T_j \leq t} Y_j, \quad X_0 = x \geq 0. \quad (1)$$

Theorem 1 (Exact affine Markov closure). *For $t, s \geq 0$, with $\alpha = \kappa/\gamma$,*

$$X_t = e^{-\gamma t} x + \sum_{T_j \leq t} e^{-\gamma(t-T_j)} Y_j, \quad (2)$$

$$\mathbb{E}_x e^{-sX_t} = e^{-se^{-\gamma t} x} \left(\frac{\beta + se^{-\gamma t}}{\beta + s} \right)^\alpha. \quad (3)$$

The unique invariant probability is Gamma with shape α and rate β . For every $p \geq 1$,

$$W_p(P_t(x, \cdot), P_t(y, \cdot)) = e^{-\gamma t} |x - y|. \quad (4)$$

In stationarity,

$$\mathbb{E} X^n = \frac{(\alpha)_n}{\beta^n}, \quad \text{cum}_n(X) = \frac{\alpha(n-1)!}{\beta^n}, \quad \text{Cov}(X_t, X_0) = \frac{\alpha}{\beta^2} e^{-\gamma|t|}. \quad (5)$$

Finally, the generator preserves $\mathcal{P}_m = \text{span}\{1, x, \dots, x^m\}$, and its restriction has exactly the simple eigenvalues $0, -\gamma, \dots, -m\gamma$. This last assertion concerns each fixed finite-dimensional filtration only; it is not a statement about the full L^2 spectrum, completeness, normality, or reversibility.

Proof of the path and transform. Variation of constants between successive jumps proves (2). The Laplace functional of a marked Poisson process gives

$$\begin{aligned} \log \mathbb{E} \exp \left[-s \sum_{T_j \leq t} e^{-\gamma(t-T_j)} Y_j \right] &= \kappa \int_0^t \left(\frac{\beta}{\beta + se^{-\gamma u}} - 1 \right) du \\ &= \frac{\kappa}{\gamma} \log \frac{\beta + se^{-\gamma t}}{\beta + s}. \end{aligned}$$

Multiplication by the deterministic initial term proves (3). Factoring the right side first at decay $r_1 = e^{-\gamma t_1}$ and then at $r_2 = e^{-\gamma t_2}$ cancels the intermediate denominator and leaves $r_1 r_2$, proving the Chapman–Kolmogorov law directly. $\square$

The phrase *shot-noise Laplace semigroup owner* is the round-zero certificate. Formula (3), not a simulation, owns the transition law.

2 Equilibrium, contraction, and correlations

Sending $t \rightarrow \infty$ in (3) gives

$$\left(\frac{\beta}{\beta+s}\right)^\alpha,$$

the $\text{Gamma}(\alpha, \text{rate } \beta)$ transform. Substitution into the semigroup formula proves invariance. If two copies use the same marked Poisson process, (2) gives the deterministic difference $e^{-\gamma t}(x-y)$. Their laws are translates, so this coupling gives the upper bound in (4); Jensen's inequality applied to their mean difference gives the same lower bound. Any invariant law must, after its Laplace transform is propagated and $t \rightarrow \infty$, have the Gamma transform. This proves uniqueness without presupposing a moment.

Expansion of the logarithm of the Gamma transform yields the cumulants in (5); differentiation gives its rising-factorial moments. The conditional mean is

$$\mathbb{E}[X_t | X_0] = e^{-\gamma t} X_0 + \frac{\kappa}{\gamma\beta}(1 - e^{-\gamma t}).$$

After stationary centering, multiplication by X_0 gives the covariance for $t \geq 0$. Scalar stationarity supplies the displayed even extension. These identities also agree with the classical shot-noise and Lévy-driven OU sources [1, 2, 3].

3 Exact finite polynomial filtration

For suitable f , the generator is

$$Lf(x) = -\gamma x f'(x) + \kappa \int_0^\infty \beta e^{-\beta y} [f(x+y) - f(x)] dy. \quad (6)$$

Because $\mathbb{E}Y^r = r!/\beta^r$, binomial expansion gives, for every $n \geq 0$,

$$Lx^n = -n\gamma x^n + \kappa \sum_{j=0}^{n-1} \binom{n}{j} \frac{(n-j)!}{\beta^{n-j}} x^j. \quad (7)$$

Thus $L\mathcal{P}_m \subseteq \mathcal{P}_m$. In the ordered monomial basis, the matrix is triangular with pairwise distinct diagonal $0, -\gamma, \dots, -m\gamma$. It is therefore diagonalizable, every listed eigenvalue is simple, and backward substitution gives one monic eigenpolynomial of each degree. This is a complete finite-dimensional statement for every m ; taking an unsupported limit in m would not prove a full L^2 spectral theorem, and we do not take it.

4 Degenerate faces and Route-A boundary

If $\kappa = 0$ but $\gamma > 0$, the process is deterministic decay and its unique invariant probability is δ_0 . If $\gamma = 0 < \kappa$, it is a nondecreasing compound-Poisson subordinator and has no invariant probability. At $\gamma = \kappa = 0$ it is static and every initial law is invariant. The value $\beta = 0$ is not an exponential probability law; $\beta \rightarrow \infty$ is only a weak zero-mark limit. These faces are not silently inserted into formulas containing κ/γ .

The finite certificate records exact Gamma moments and generator matrices through degree 12, sixty transition values, and ten semigroup products. Independent and symbolic implementations recompute them. These rows are regression evidence, not the proof of an infinite-time theorem.

C229 is a square-root diffusion, C233 is a discrete immigration-death chain, C265 has self-exciting arrivals, and C328 has dichotomous velocities and compact support; none owns the present positive-jump Gamma semigroup. All five Route-A layers fail: there is no intrinsic prime owner, isolated periodic ledger, orbit determinant, target analytic bridge, or natural self-adjoint target lift. Route B is not invoked. No target local data, Euler factors, root numbers, automorphy, target divisor law, functional equation, zero match, or Hilbert-Pólya operator is claimed.

References

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